

A Tower-Recursive Source for Every Gold Diagonal

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Abstract

Let m be a power of two, let $K_m = \mathbb{F}_{2^m}$ be the canonical finite nim-field, and write $e_i = 2^i$ for its Conway bit basis. For the Gold form

$$Q_a(x) = \text{Tr}_{K_m/\mathbb{F}_2}(x^{1+2^a}),$$

the diagonal constants $Q_a(e_i)$ are independent of the polar form. We give a recursion for the trace-dual diagonal of every scaled Gold component, prove that the unscaled dual always descends to the half-field, and conclude that it is an Artin–Schreier value $w^2 + w$. Thus every Gold exponent has a single tower-constructed source w satisfying

$$\text{Tr}((w^2 + w)e_i) = Q_a(e_i) \quad (0 \leq i < m).$$

The construction works uniformly for odd and even a . It supplies the algebraic source used by the companion weighted-source Witt–FIFO theorem, while leaving the strategic normal-play argument logically separate.

1 Introduction and main theorem

For a scale $c \in K_m$, define the scaled Gold component

$$Q_{a,c}^{(m)}(x) = \text{Tr}_m(cx \text{Fr}^a(x)), \quad \text{Fr}(x) = x^2,$$

where $\text{Tr}_m = \text{Tr}_{K_m/\mathbb{F}_2}$. Its canonical-basis diagonal extends uniquely to an $\mathbb{F}_2$ -linear functional

$$\ell_{a,c}^{(m)}\left(\sum_i x_i e_i\right) = \sum_i x_i Q_{a,c}^{(m)}(e_i).$$

The absolute trace pairing on a finite separable field is nondegenerate, so there is a unique $\Lambda_{a,c}^{(m)} \in K_m$ such that

$$\text{Tr}_m(\Lambda_{a,c}^{(m)} e_i) = \text{Tr}_m(c e_i \text{Fr}^a(e_i)) \quad \text{for every } i < m. \quad (1)$$

Write $\Lambda_a^{(m)} = \Lambda_{a,1}^{(m)}$.

Theorem 1.1 (all-exponent Gold-diagonal source). *For every power of two $m \geq 2$ and every $a \geq 0$:*

1. $\Lambda_{a,c}^{(m)}$ is given by the quadratic-tower recursion in Theorem 2.2, using only nim addition and multiplication, Frobenius, relative trace, and the canonical tower generator;
2. the unscaled dual descends:

$$\Lambda_a^{(m)} \in K_{m/2} \subset K_m;$$

3. there is $w_a^{(m)} \in K_m$ with

$$\Lambda_a^{(m)} = \wp(w_a^{(m)}) = (w_a^{(m)})^2 + w_a^{(m)};$$

and therefore

$$Q_a(e_i) = \text{Tr}_m(\wp(w_a^{(m)})e_i) \quad (0 \leq i < m).$$

The two possible sources are $w_a^{(m)}$ and $w_a^{(m)} + 1$.

The theorem is about sourcing the diagonal constants. It does not itself infer a P -set theorem or a linking strategy. Those are supplied by the separate weighted-source Witt–FIFO theorem; this proof/semantics boundary is essential even though the companion normal-play construction supplies the remaining game-semantic layer.

2 The quadratic nim-tower block

Put $m = 2M$. The canonical nim tower has

$$K_m = K_M(u) = K_M \oplus uK_M, \quad u = e_M,$$

and the nontrivial K_M -automorphism $\sigma = \text{Fr}^M$ satisfies

$$\sigma(u) = u + 1.$$

The canonical basis splits as

$$e_0, \dots, e_{M-1}, \quad ue_0, \dots, ue_{M-1};$$

indeed $ue_j = e_{M+j}$. Let

$$T = T_{m/M} = 1 + \sigma : K_m \longrightarrow K_M$$

be the relative trace.

Lemma 2.1 (trace-pairing blocks). *For $A, B, e \in K_M$,*

$$T((A + uB)e) = Be, \tag{2}$$

$$T((A + uB)ue) = (A + B)e. \tag{3}$$

Proof. Because σ fixes A, B, e and sends u to $u + 1$,

$$(A + uB)e + \sigma((A + uB)e) = (A + uB)e + (A + (u + 1)B)e = Be.$$

For the upper block, apply the same calculation to $(A + uB)u$:

$$\begin{aligned} & (A + uB)u + (A + (u + 1)B)(u + 1) \\ &= Au + u^2B + Au + A + (u + 1)^2B = A + B, \end{aligned}$$

since characteristic two cancels the doubled Au , u^2B , and uB terms. Multiplication by e gives (3). $\square$

Now write $c = c_0 + uc_1$ and set

$$d_0 = T(c), \quad d_1 = T(cu \operatorname{Fr}^a(u)). \quad (4)$$

For a lower basis vector $e \in K_M$,

$$Q_{a,c}^{(m)}(e) = \operatorname{Tr}_M(T(c)e \operatorname{Fr}^a(e)) = Q_{a,d_0}^{(M)}(e).$$

For its upper mate ue ,

$$Q_{a,c}^{(m)}(ue) = \operatorname{Tr}_M(T(cu \operatorname{Fr}^a(u))e \operatorname{Fr}^a(e)) = Q_{a,d_1}^{(M)}(e).$$

Thus d_0 and d_1 are exactly the smaller-field scales of the two diagonal blocks.

Theorem 2.2 (scaled dual recursion). *Let*

$$L_0 = \Lambda_{a,d_0}^{(M)}, \quad L_1 = \Lambda_{a,d_1}^{(M)},$$

with d_0, d_1 as in (4). Then

$$\Lambda_{a,c}^{(2M)} = (L_0 + L_1) + uL_0. \quad (5)$$

The base case over $K_1 = \mathbb{F}_2$ is $\Lambda_{a,c}^{(1)} = c$.

Proof. Write the candidate in (5) as $A + uB$, where $B = L_0$ and $A = L_0 + L_1$. By Lemma 2.1, its relative trace pairing with a lower basis vector e is L_0e , while its pairing with ue is

$$(A + B)e = (L_0 + L_1 + L_0)e = L_1e.$$

After applying Tr_M , these are exactly the two defining dualities for the lower and upper diagonal blocks. Nondegeneracy of the absolute trace pairing makes the candidate unique, proving (5). $\square$

The scale d_1 may change at every tower level; the recursion transports that change rather than requiring one fixed dual element throughout the tower.

3 Descent and the Artin–Schreier source

Lemma 3.1 (unscaled half-field descent). *For $m = 2M$,*

$$\Lambda_a^{(m)} \in K_M.$$

Proof. In the unscaled case $c = 1$, the lower scale is

$$d_0 = T(1) = 1 + 1 = 0.$$

Linearity (or the recursion from the base case) gives $L_0 = \Lambda_{a,0}^{(M)} = 0$. Theorem 2.2 therefore reduces to

$$\Lambda_a^{(2M)} = L_1 \in K_M.$$

$\square$

Lemma 3.2 (Artin–Schreier exact sequence). *For a finite field $K = \mathbb{F}_{2^m}$,*

$$\operatorname{im}(x \mapsto x^2 + x) = \ker(\operatorname{Tr}_{K/\mathbb{F}_2}).$$

Each element of this common hyperplane has exactly two preimages, differing by 1.

Proof. The Artin–Schreier map $\wp(x) = x^2 + x$ is $\mathbb{F}_2$ -linear and has kernel $\mathbb{F}_2 = \{0, 1\}$, so its image has $|K|/2$ elements. Frobenius invariance of trace gives

$$\mathrm{Tr}(\wp(x)) = \mathrm{Tr}(x^2) + \mathrm{Tr}(x) = \mathrm{Tr}(x)^2 + \mathrm{Tr}(x) = 0,$$

so the image lies in the trace-zero hyperplane. The finite-field trace is a nonzero $\mathbb{F}_2$ -linear functional, hence its kernel also has $|K|/2$ elements. The inclusion is equality. Every nonempty fibre is a coset of the two-element kernel. $\square$

Proof of Theorem 1.1. Theorem 2.2 proves the constructive first assertion and Lemma 3.1 proves the second. If $\lambda = \Lambda_a^{(2M)} \in K_M$, trace transitivity gives

$$\mathrm{Tr}_{2M}(\lambda) = \mathrm{Tr}_M(\mathrm{Tr}_{K_{2M}/K_M}(\lambda)) = \mathrm{Tr}_M(2\lambda) = 0.$$

Lemma 3.2 supplies w with $w^2 + w = \lambda$. Substituting this identity into the defining trace duality (1) yields the stated formula for every basis coin. The fibre statement in Lemma 3.2 gives the two sources w and $w + 1$. $\square$

4 Implementation and formal boundary

The Rust core implements the proof literally:

- `gold_component_diagonal_dual(m,a,c)` evaluates Theorem 2.2;
- `gold_diagonal_dual(m,a)` specializes to $c = 1$; and
- `gold_diagonal_artin_schreier_source(m,a)` uses the existing exact finite-nim-field solver to choose one of the two sources.

The implementation is checked directly against all defining trace-pairing equations for every scale over K_m at $m \leq 8$. For the unscaled theorem it checks half-field descent, the Artin–Schreier equation, and every diagonal coordinate at representative exponents through the full $m = 128$ backend. The previous even- a values

$$\Lambda_2^{(4)} = 0, \quad \Lambda_2^{(8)} = 6, \quad \Lambda_2^{(16)} = 102, \quad \Lambda_2^{(32)} = 24582$$

and $\Lambda_4^{(16)} = 31$, $\Lambda_4^{(32)} = 8030$ are pinned as regression values; their drift is now explained and handled by the scale recursion.

The Lean module `formal/Ogdoad/GoldDiagonal.lean` kernel-checks the quadratic trace-block formulas, the dual-block reconstruction, the vanishing of the absolute trace of a quadratic-extension base element, additivity of the Artin–Schreier map, the one-step tower lift of a source, and the full finite-field equivalence $\mathrm{Tr}(\lambda) = 0 \iff \lambda = w^2 + w$. The last result is proved from rank-nullity, trace surjectivity, and the two-element kernel, rather than introduced as an axiom. The module does not encode the finite nim tower or replace the Rust implementation with a Lean finite-field model.

5 Integration with Gold normal play

For every exponent, the diagonal is supplied by one deterministically selected tower constant through nim multiplication, Frobenius, trace, and finally one square plus XOR at the source identity $\Lambda = \wp(w)$. No coordinate-by-coordinate evaluation of Q_a occurs in the recursion. Unlike the $a = 1$ Fermat-coin formula, the final w is selected by exact finite-field linear algebra; the theorem does not claim a comparably simple closed expression or that this selection itself satisfies the strategic naturality axioms.

The strategic layer is supplied separately by weighted-source Witt-FIFO. Its source-pair weights consume these diagonal bits locally, the matching theorem forces the exact charge in every dimension, and the phase-aware claim lantern gives an ordinary normal-play P-set $\{Q_a = 0\}$. The present theorem supplies the all- a algebraic input to that construction; it does not collapse the distinction between sourcing and play semantics.